

Project ID :

25-26J-091

1. Topic (12 words max)

Smart Real-Time Age and Emotion-Based Advertising in Malls Using Custom-Built Models

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

In Sri Lankan shopping malls, digital advertisements are currently static or follow a fixed schedule, regardless of the audience present at any given time. This often results in a disconnect between the viewers and the content displayed, reducing engagement and advertising impact. Modern consumers are increasingly exposed to digital content that is interactive, personalized, and emotionally resonant. However, local shopping malls have yet to adopt such intelligent systems that adapt advertisements based on real-time audience demographics and mood.

This research aims to bridge that gap by introducing a real-time, AI-powered advertising system that uses computer vision and emotion detection to analyze the age and emotional state of mall visitors. The proposed system utilizes existing CCTV infrastructure and image datasets to extract demographic data, such as estimated age groups and facial expressions (happy, neutral, sad), to determine the mood of the majority audience present. Using this data, the system dynamically selects and displays the most relevant advertisements from a categorized database, ensuring a better match between viewer interest and advertisement content.

By making advertisements contextually relevant, the system has the potential to improve viewer satisfaction and brand engagement. It also encourages the integration of AI technologies into local retail infrastructure. This project will use image processing, age classification, emotion detection models, and sentiment analysis to optimize advertisement display. The solution also addresses privacy concerns by avoiding personal identification and focusing only on statistical data. This research will benefit advertisers, mall operators, and customers alike by making digital marketing more intelligent, targeted, and impactful.

- Y. Zhang, D. Zhou, S. Chen, S. Gao, and Y. Ma, “Single-Image Crowd Counting via Multi-Column Convolutional Neural Network,” *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 589–597.
- V. A. Sindagi and V. M. Patel, “A Survey of Recent Advances in CNN-Based Single Image Crowd Counting and Density Estimation,” *Pattern Recognition Letters*, vol. 107, pp. 3–16, 2018.
- W. Wang, J. Shen, and L. Shao, “Consistent Video Saliency Using Local Gradient Flow Optimization and Global Refinement,” *IEEE Transactions on Image Processing*, vol. 24, no. 11, pp. 4185–4196, Nov. 2015.
- J. Redmon and A. Farhadi, “YOLOv3: An Incremental Improvement,” arXiv preprint arXiv:1804.02767, 2018.
- A. Mollahosseini, D. Chan, and M. H. Mahoor, “Going Deeper in Facial Expression Recognition Using Deep Neural Networks,” *2016 IEEE Winter Conference on Applications of Computer Vision (WACV)*, Lake Placid, NY, 2016, pp. 1–10.
- Z. Yu and C. Zhang, “Image Based Static Facial Expression Recognition with Multiple Deep Network Learning,” *Proceedings of the ACM International Conference on Multimodal Interaction*, 2015, pp. 435–442.
- X. Wang, M. Yang, S. Zhu, and Y. Lin, “Regionlets for Generic Object Detection,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 37, no. 10, pp. 2071–2084, Oct. 2015.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed system integrates custom-built computer vision models and intelligent decision logic to dynamically adapt digital advertisements in Sri Lankan shopping malls. It consists of four core components: (1) Age Classification, (2) Emotion Classification, (3) Crowd Detection, (4) Advertisement Categorization and Personalization Engine.

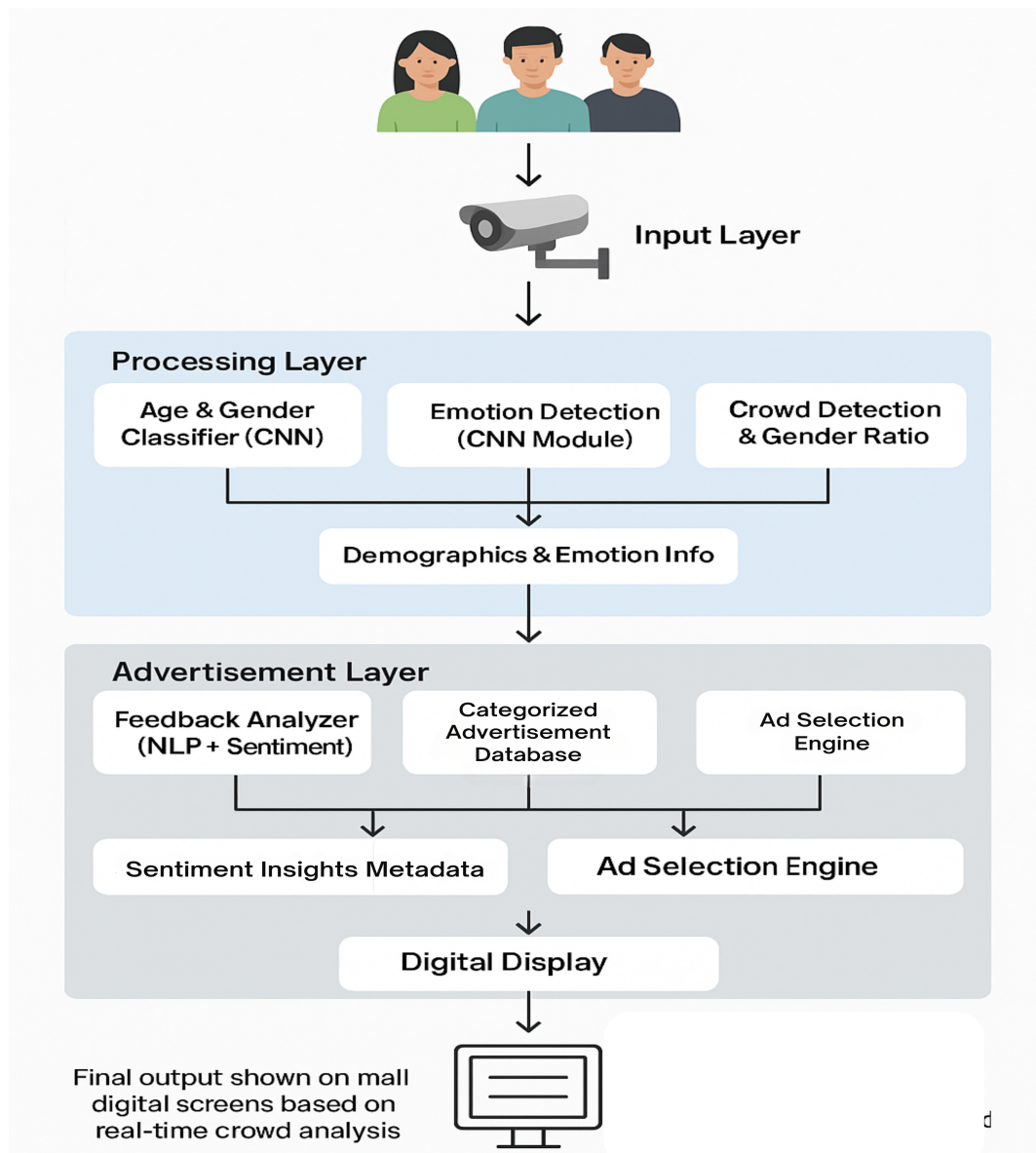
Unlike many existing solutions that rely on pre-trained models, this project will develop **age and emotion classification models from scratch** using raw image datasets. These models will be trained on cleaned, annotated image datasets. Deep learning techniques such as CNNs will be used to detect and classify facial features into appropriate categories (e.g., child, adult, senior; happy, neutral, sad).

Crowd detection will utilize live video streams to estimate crowd size in front of digital displays. Techniques like background subtraction and person-detection using object tracking algorithms will be developed to identify how many people are present at a given time.

The categorized advertisement database will include metadata like suitable age group, emotional appeal, and crowd suitability. Based on the real-time demographic and emotional profile, the system dynamically selects the most relevant ad to display. Feedback from social media or surveys will be optionally analyzed using NLP to improve ad targeting accuracy.

This end-to-end, customized solution focuses on context-aware ad delivery by leveraging real-time image and video analytics, while also addressing data privacy and ethical considerations.

Concept Diagram:



7. Brief description of specialized domain expertise, knowledge, and data requirements
(300 words max)

Specialized Domain Expertise:

- **Deep Learning & Computer Vision:**
Designing and training age and emotion classification models **from scratch** using convolutional neural networks (CNNs). This requires expertise in neural network architecture design, data preprocessing, augmentation, training, validation, and optimization.
- **Crowd Analysis & Object Detection:**
Building video-based crowd detection using **classical image processing** and object detection. Expertise is needed in methods like background subtraction, frame differencing, HOG-based detection, and real-time image analysis using OpenCV or custom-built algorithms.
- **NLP & Sentiment Analysis:**
Applying natural language processing techniques such as sentiment analysis on survey responses and social media feedback. This involves knowledge of tokenization, classification, and training simple NLP models or using rule-based analyzers.

Knowledge Requirements:

- **Model Development Workflow:**
Understanding of the full model lifecycle — data handling, training, tuning, evaluation, and integration into applications.
- **Video & Image Stream Processing:**
Ability to handle frame extraction from video streams and batch process them for crowd detection and facial analysis.
- **Ad Recommendation Logic:**
Knowledge of designing rule-based or metadata-driven recommendation systems, and mapping real-world conditions (age/emotion/crowd) to business logic (ad targeting).
- **Ethical & Privacy-Aware AI Development:**
Awareness of ethical handling of visual data and building systems that analyze demographics without storing personal identifiers.

Data Requirements:

- **Raw video footages** collected from the mall.

8. Objectives and Novelty

Main Objective To develop a smart, AI-driven advertising system that dynamically selects and displays relevant advertisements in shopping malls based on real-time age group , emotional state , and crowd size of the audience, while ensuring privacy, relevance, and improved engagement. This system aims to bridge the gap between traditional digital advertising and real-time audience engagement by introducing adaptive content delivery using live visual inputs.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Fonseka W S M IT22109712	To develop an age detection model that gives different weights to facial regions to improve accuracy under real-world conditions. To build a gender classification model using optimized facial features.	• Develop a lightweight, camera-integrated interactive game (e.g., Snake game) to attract individuals to the ad screen. • Capture facial data of the player during the game for demographic analysis. • Build an age detection model that gives different weights to facial regions (e.g., eyes: high weight, hair: low weight, background: zero). • Incorporate occlusion handling techniques for masked faces or sunglasses. • Implement a gender classification model based on facial features. Generate and store face embeddings with regional weighting instead of	• Introduces weighted-region face processing in both age detection and face embedding, enhancing accuracy and robustness under occlusions. • Privacy-preserving face data handling by avoiding raw image storage and using only embedding vectors and coordinates. • Uses gamification to drive voluntary engagement, encouraging users to look at the camera naturally. • Dual-level advertising model: individual-level demographic collection through gameplay and crowd-based ad recommendations immediately afterward.

		raw images. Store facial coordinate vectors to enable returning user identification for the game leaderboard. After gameplay, trigger a 5-minute ad display targeting the mass audience based on majority demographic detection.	Enables non-intrusive identification of returning users using face embeddings without violating privacy or needing logins.
Rajana H T R IT22071934	To develop a personalized advertisement recommendation system using collaborative, content-based, hybrid, and demographic filtering methods. To integrate outputs from real-time age, emotion, and behavioral detection models into the recommendation logic.	- Implement collaborative filtering using user interaction history and inferred demographic clusters. - Apply content-based filtering based on ad metadata (category, tone, product type) matched to detected user preferences. - Combine both methods into a hybrid model for improved performance and accuracy. - Utilize demographic filtering using data from age, gender, and group behavior detection models. - Introduce serendipity by incorporating diversity-aware filtering to occasionally present novel or unexpected but relevant ads. - Add a context-aware layer that adapts recommendations based on time, location, crowd type, and past engagement levels. - Mitigate the cold start problem by using group-level behavior and content metadata to generate	Serendipity-focused ad recommendation: Introduces unexpected but enjoyable ads, increasing user engagement and advertiser value. Context-aware layer allows the system to adapt dynamically to external factors like time of day, current crowd composition, and engagement levels. Diversity-aware filtering prevents repetition and ad fatigue while maintaining relevance. Solves the cold start problem by leveraging demographic and behavior detection outputs for users without prior interaction history. - Creates a smarter and more emotionally resonant advertising experience that is both personalized and surprising, which enhances recall and impact.

		recommendations even for first-time users.	
Samaraweera W D U I IT22258526	To create an emotion detection system that works in real-time CCTV footage, accurately identifying faces under occlusion (such as masks or caps), and classifying domain-specific emotions like curious or suspicious using a mix of culturally relevant and occlusion-handling datasets, to trigger adaptive advertisements and raise security alerts in shopping malls.	• Build a system that can detect people's emotions in real-time by analyzing live video from CCTV cameras. • Ensure the system works even when faces are partially covered by masks or caps, as commonly seen in public areas like shopping malls. • Use machine learning models to detect specific emotions such as: ▫ Happy, Sad, Angry, Neutral → used for triggering emotion-based digital advertisements ▫ Suspicious or ignoring → used for raising security alerts • Design the system to filter these predicted emotions into two separate outputs: ▫ One for advertising purposes ▫ Another for security alerting • Ensure user privacy by not storing any images or personal data — only the detected emotion labels are used for action.	This is a combination of real-time emotion detection, intelligent action filtering, and privacy-aware deployment — all in one unified system. • Detects both basic emotions (e.g., happy, sad, angry, neutral) for advertising and domain-specific emotions (e.g., suspicious, ignoring) for security, even when faces are masked or partially occluded. • Filters predicted emotions into two action paths — one for playing targeted advertisements based on emotional state, and the other for raising real-time security alerts using a rule-based classification technique.
Abeykoon S N IT22184030	To design and implement a real-time system that captures individuals from CCTV footage, extracts key pose features (such as	• Build a system that can detect user engagement in real-time by analyzing live video from CCTV cameras.	This is the first known approach in mall-based advertising that uses a combination of body pose orientation and dwell time—without any facial recognition—to detect and quantify real-time audience engagement.

	head, shoulder, and torso orientation), and analyzing their body orientation, stillness and calculates dwell time to detect and quantify engagement levels of individuals in public retail spaces (such as malls) without using facial data and feeding engagement data into a recommender system for adaptive ad optimization.	• Detect and track multiple individuals simultaneously within crowded retail environments. • Use pose estimation techniques to extract key body landmarks such as head, shoulders, and torso. Normalize pose data for consistent analysis across different people and camera angles. • Calculate orientation (e.g., angle of body facing the screen) from pose landmarks. • Measure dwell time by counting continuous frames a person remains stationary in front of the screen. • Design an algorithm to assign an engagement score based on body orientation and dwell time. Define engagement thresholds to classify individuals as engaged or not engaged. • Provide real-time insights to support adaptive advertising strategies and optimal screen placements. • Extend the system to detect anomalous or suspicious behaviors using pose-based patterns, contributing to both marketing analytics and retail security.	• Behavior-Only Engagement Detection: Measures engagement strictly through body pose and movement cues, notably orientation and dwell time - no face or personal data needed. • Introduces a new Engagement Index (E-Index) that combines pose direction and dwell duration to quantitatively measure attention toward an ad. • Real-Time, Continuous Analytics in Crowds: Generates ongoing engagement metrics for everyone, robustly operating even in dynamic, crowded retail environments—not just in static or controlled settings.
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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Fonseka W S M IT22109712	This component directly applies core Data Science principles using machine learning and computer vision to extract demographic insights from facial images. It focuses on building supervised models for age and gender classification using labeled image datasets. An important aspect of this model is the use of feature weighting—assigning more importance to key facial regions such as the eyes and nose, while giving less weight to areas like hair or background. This technique enhances prediction accuracy, especially in real-world scenarios where faces may be partially covered by masks or sunglasses. The component also introduces dimensionality reduction through face embeddings. These embeddings convert high-dimensional facial images into low-dimensional vectors, which are efficient for storage and comparison. This allows the system to recognize returning users without storing raw images. Moreover, the use of embedding vectors instead of raw image data supports privacy-preserving data handling. This approach aligns with ethical practices in data science, addressing concerns related to user privacy and compliance with data protection standards. Overall, this component showcases the practical use of Data Science in real-time applications by integrating supervised learning, feature engineering, and ethical data management.
Rajana H T R IT22071934	This component utilizes multiple recommendation techniques—collaborative filtering, content-based filtering, and hybrid models—which are all fundamental topics in Data Science. These methods allow the system to personalize ad delivery based on both user similarity and content relevance. It also incorporates demographic filtering, using outputs from age, gender, and emotion detection models to enhance the recommendation logic. This results in a system that not only adapts to individuals but also reflects crowd-level trends and preferences. To address the cold start problem, the system uses demographic and behavioral data when user history is unavailable. This enables effective ad targeting even for first-time or anonymous users. One of the unique features of this component is the inclusion of serendipity—deliberately recommending unexpected but enjoyable ads. This is achieved through diversity-aware filtering and a context-aware layer that considers environmental factors like time of day, audience type, or prior engagement. These enhancements make the ad recommendation process smarter and more engaging. By combining real-time data processing, behavioral analytics, and personalization, this component demonstrates a strong alignment with the capabilities and objectives of a Data Science specialization.

Samaraweera W D U I IT22258526	The emotion detection component aligns directly with the Data Science specialization by applying supervised deep learning to solve a real-world facial expression classification task. It follows the full data science workflow, including data selection, preprocessing, model design, training, validation, and evaluation. A custom convolutional neural network (CNN) is developed and trained on the Facial Expression Dataset – Sri Lankan, ensuring cultural relevance and local adaptability. The emotion filtering and decision-making component aligns strongly with the Data Science specialization by applying rule-based classification techniques to map multi-class emotion outputs into binary action categories — targeted advertisements or security alerts. Key skills demonstrated include CNN architecture design, data augmentation, hyperparameter tuning, and loss optimization. Performance is evaluated using metrics such as accuracy, precision, recall, and confusion matrices, enabling data-driven refinement. The component also tackles practical challenges like image noise, class imbalance, and generalization, showing how deep learning can support intelligent, real-time emotion recognition in surveillance systems.
Abeykoon S N IT22184030	The engagement detection component aligns directly with the Data Science specialization by applying real-time pose analysis and behavioral modeling to a practical retail analytics challenge. It follows the full data science workflow, including data collection from surveillance feeds, preprocessing, feature extraction, model formulation, threshold calibration, and visualization of results. The system uses lightweight pose estimation models to extract key posture features such as head and shoulder alignment and stationary duration. These features are engineered into quantifiable engagement scores that drive downstream analytics. Key skills demonstrated include video frame processing, pose-based feature engineering, behavioral metric modeling, real-time data pipeline development, and evaluation using temporal and spatial heatmaps. Engagement patterns are validated through aggregated metrics such as dwell time distributions and spatial crowd behavior maps. This component also addresses real-world complexities such as occlusion, partial visibility, and non-cooperative subjects, illustrating how behavioral-based machine learning techniques can drive intelligent, privacy-preserving retail analytics in live environments.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Dr	Nathali	Silva	
Co-Supervisor	Ms	Shashini	Kumarasinghe	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.